

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 48 and 49

The line $x + 2y + a = 0$ intersect the circle $x^2 + y^2 = 4$ at points A and B. Another line $12x - 6y - 41 = 0$ intersect the circle $x^2 + y^2 - 4x - 2y + 1 = 0$ at two distinct points C and D. Let another circle $S = 0$ is passing through the points A, B, C and D

48. If the equation of circle $S = 0$ is $x^2 + y^2 + 2gx + 2fy + c = 0$, then $|2g + 2f + c|$ is
49. Values of a is

Paragraph for Question Nos. 50 and 51

Given $f(x + y + z) = f(x) f(y) f(z) \forall x, y, z \in \mathbb{R}$ and $f(2) = 4$ and $f'(0) = 3$

50. Number of possible value(s) of $f(0)$ is/are
51. Sum of digits of $f'(2)$ is